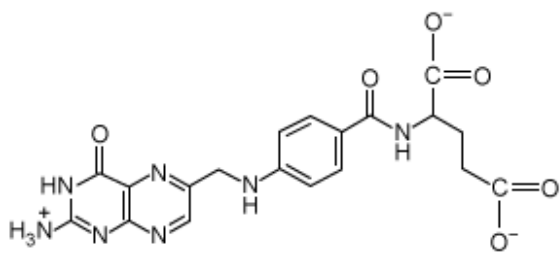


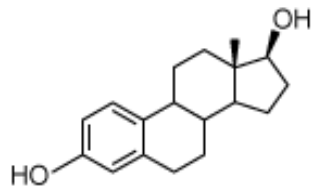
Which of the following molecules could participate in cell signaling without a second messenger?

☐ A.



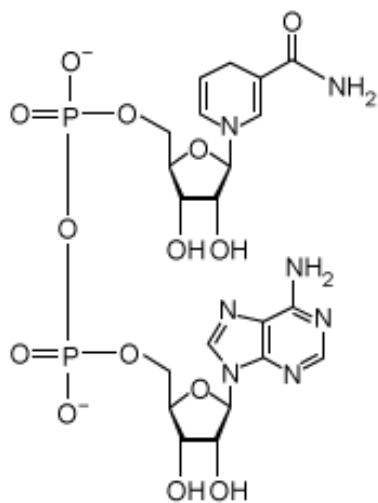
[6%]

✓ ☒ B.



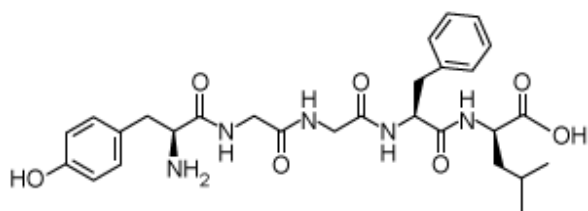
[74%]

☐ C.



[12%]

☐ D.



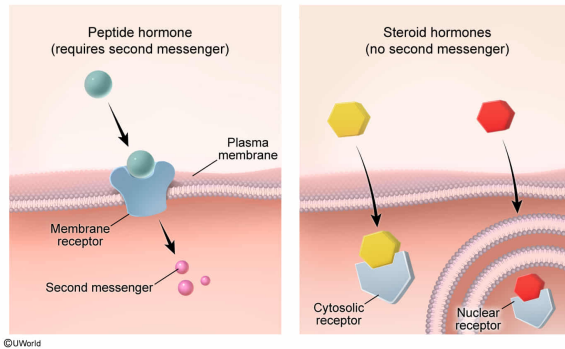
[6%]

Correct

74% answered correctly

Explanation:

Peptide hormones bind membrane receptors;
steroid hormones diffuse through the membrane



Cells communicate with and respond to their environments through **chemical signals**. Chemical signals must be transmitted across the cell membrane to induce a response. Hydrophilic signal molecules such as **peptide hormones** interact poorly with the hydrophobic tails of **phospholipid bilayers**, so they **cannot easily cross cell membranes**. Instead, these hormones (first messengers) bind to cell surface receptors such as **G protein–coupled receptors**. These receptors relay the signal from *extracellular ligands* by causing the production of new molecules, called **second messengers**, on the *cytosolic side* of the membrane. The second messenger molecules can then interact with proteins in the cytosol and nucleus, leading to a response.

In contrast, hydrophobic molecules such as **steroid hormones** can cross the cell membrane and directly interact with effector proteins inside the cell. Therefore, these hormones **do not require second messengers**. Steroid hormones are derived from cholesterol and are easily recognized by their **characteristic backbone structure**, consisting of three 6-membered rings and one 5-membered ring fused together. Estradiol, the molecule shown in Choice B, contains this characteristic backbone and is an example of a steroid hormone that signals without a second messenger.

(Choice A) This molecule is folic acid, also known as vitamin B₉, which is not typically involved in signaling. The multiple nitrogen and oxygen atoms in the structure, as well as the multiple charged functional groups, make it hydrophilic and unable to readily cross the cell membrane. If it did signal, it would require a second messenger.

(Choice C) This molecule is the electron carrier NADH. The charged phosphate groups make it hydrophilic, so if NADH were involved in cell signaling, it would require a second messenger.

(Choice D) This structure is a polypeptide, and therefore it would act as a peptide hormone in signaling. Peptide hormones require second messengers.

Educational objective:

Cells communicate through chemical signals called hormones. Hydrophilic molecules such as peptide hormones cannot cross the cell membrane and require second messengers. Hydrophobic molecules such as steroid hormones can cross the cell membrane and do not need second messengers.